UI/UX PATTERNS → LML/NSL INTEGRATION

Piping 343 Interface Patterns Through Universal Compiler

Input: 343 UI/UX patterns from EFUE collapse
Pipeline: UIC → UUIS → LML + NSL
Output: Executable interface laws + usability repair moves

UIC: UNIVERSAL IDENTITY COMPILER

Input Artifact:

343 Interface Pattern Classes (9 dimensions × 9 levels)

UIC Step 1: Extract Primitives

State Variables (x):

- Current interface signature: [U1.x, U2.x, U3.x, U4.x, U5.x, U6.x, U7.x, U8.x, U9.x]
- Information density: Elements visible
- Interaction method: User action types
- Feedback timing: Response latency
- Error states: Current error conditions
- User proficiency level: Novice/intermediate/expert
- Device context: Mobile/desktop/terminal
- Task being performed: Current user goal

Invariants (I):

1. **Cognitive Load Limit:** MUST NOT exceed human working memory capacity
    - “Information density (U1.x) MUST respect 7±2 item limit for untrained users”
    - “If U1.4+ (rich/dense) THEN user MUST be expert OR have training”
2. **Feedback Responsiveness:** Interactive elements MUST provide feedback within

tolerance

   - "U2.2-U2.3 (direct/single-click) MUST have U3.0-U3.2 (instant-responsive) feedback"
   - "Feedback delay MUST NOT exceed user patience threshold for task type"
3. **Error Recoverability:** Critical operations MUST have error protection
   - "Destructive actions MUST have U4.2 (confirmation) OR U4.3 (undo)"
   - "CANNOT have U4.7-U4.8 (cryptic/catastrophic) errors in production"
4. **Learning Curve Appropriateness:** Complexity MUST match target user
   - "Consumer apps MUST be U5.0-U5.3 (obvious to tutorial-assisted)"
   - "Professional tools MAY be U5.4-U5.6 IF productivity gains justify"
5. **Navigation Clarity:** User MUST always know where they are and how to return
   - "Navigation depth MUST have visible breadcrumbs OR back button"
   - "Modal layers MUST have clear exit path"
6. **Visual Hierarchy Requirement:** Important elements MUST be visually distinct
   - "Primary actions MUST be U7.3+ (size/color/depth hierarchy)"
   - "CANNOT be U7.0 (completely flat) unless intentionally minimal"
7. **Interaction Consistency:** Paradigm MUST be consistent throughout interface
   - "CANNOT mix U8.8 (CLI) with U8.2 (point-click) without clear mode indication"
   - "Gestures MUST be platform-standard OR explicitly taught"
8. **Dimensional Coherence:** All 9 dimensions MUST be mutually compatible
   - "U1.7+ (overwhelming) incompatible with U5.0 (instantly obvious)"
   - "U2.8 (programming) requires U5.7 (mastery) AND U8.8 (CLI)"

Disturbances (Δ):

- Cognitive overload: Too much information
- Response lag: Slow feedback
- Error frustration: Can't recover
- Navigation confusion: Lost in interface

- Learning abandonment: Too hard to learn
- Inconsistency: Mixed paradigms
- Pattern violation: Breaks user expectations
- Accessibility failure: Unusable for some users

Couplings (A/B):

- Density ↔ Learning curve (dense requires training)
- Interaction ↔ Feedback (direct manipulation needs instant feedback)
- Error tolerance ↔ User type (novices need more protection)
- Navigation ↔ Density (complex content needs clear navigation)

UIC Output (PER):

```
PER_UI_UX = {
  x: {
    signature: [U1.x, U2.x, U3.x, U4.x, U5.x, U6.x, U7.x, U8.x, U9.x],
    visible_elements: count,
    interaction_state: current_mode,
    feedback_latency: measured_ms,
    error_state: current_errors,
    user_proficiency: estimated_level,
    device: context,
    task: current_goal
  },

  I: [
    "Cognitive load MUST NOT exceed capacity",
    "Feedback MUST be responsive",
    "Errors MUST be recoverable",
    "Learning curve MUST match user",
    "Navigation MUST be clear",
    "Visual hierarchy MUST exist",
    "Interaction MUST be consistent",
    "Dimensions MUST be coherent"
  ],

  Δ: [
    "Cognitive overload",
    "Response lag",
    "Error frustration",
    "Navigation confusion",
```

```
    "Learning abandonment",
    "Inconsistency",
    "Pattern violation",
    "Accessibility failure"
  ],

  A_B: {
    couplings: [
      "Density → Learning",
      "Interaction → Feedback",
      "Error tolerance → User type",
      "Navigation → Density"
    ]
  },

  HOLES: [
    "How to measure cognitive load? Need metrics",
    "How to detect navigation confusion? Need analytics",
    "How to determine optimal feedback timing? Need user research",
    "What to do when pattern violated? Need repair strategy"
  ]
}
```

UUIS: UNIVERSAL SNAP COMPILER

Block 1: IDENTITY KERNEL

```
CORE INTERFACE INVARIANTS:

1. Cognitive Capacity Law
   "Interface MUST NOT exceed human working memory limit (7±2 items) without chun

2. Responsiveness Law
   "Interactive elements MUST provide feedback within expected timeframe for inte

3. Error Protection Law
   "Destructive/irreversible operations MUST have confirmation OR undo capability

4. Learnability Law
   "Interface complexity MUST match target user expertise level"

5. Navigation Transparency Law
   "User MUST always know current location and how to navigate"
```

6. Visual Hierarchy Law
   "Important elements MUST be visually distinguishable from less important"

7. Consistency Law
   "Interaction paradigm MUST be consistent throughout interface OR clearly mode-

8. Dimensional Coherence Law
   "All 9 dimensions MUST form valid, compatible pattern"

HOLES:
- Need cognitive load measurement
- Need user proficiency detection
- Need pattern validation rules

Block 2: LEGALITY REGION

FORBIDDEN INTERFACE PATTERNS:

COGNITIVE VIOLATIONS:
- U1.7-U1.8 (overwhelming/incomprehensible) for consumer apps
- U1.4+ (rich) without U5.4+ (training) for general users
- 50+ elements without clear grouping/hierarchy

RESPONSIVENESS VIOLATIONS:
- U2.2-U2.3 (direct/single-click) with U3.4+ (moderate delay)
- Touch interfaces with >100ms response
- Any interactive element >3s without progress indicator

ERROR HANDLING VIOLATIONS:
- U4.8 (catastrophic) errors in any production system
- Destructive actions without confirmation (U4.0-U4.2) OR undo (U4.3)
- U4.7 (cryptic) errors shown to end users

LEARNABILITY VIOLATIONS:
- U5.7-U5.8 (mastery/inaccessible) for consumer products
- U5.0 (instantly obvious) claim with U1.5+ (dense) interface
- Professional tool with no documentation (U5.5+)

NAVIGATION VIOLATIONS:
- >5 modal layers deep
- No back/escape from modal
- Navigation structure unclear (lost user)

HIERARCHY VIOLATIONS:
- U7.0 (flat) with U1.5+ (dense) - overwhelming

```
- Primary action not visually prominent
- No visual grouping with 10+ elements

CONSISTENCY VIOLATIONS:
- Mixing U8.8 (CLI) and U8.2 (point-click) without mode clarity
- Platform-non-standard gestures without tutorial
- Inconsistent interaction paradigm within single interface

DIMENSIONAL IMPOSSIBILITIES:
- U1.0 ∧ U1.8 (minimal AND incomprehensible - contradictory)
- U3.0 ∧ U3.8 (instant AND async - contradictory)
- U5.0 ∧ U5.8 (obvious AND inaccessible - contradictory)

ADMISSIBLE:
- All 343 core patterns
- Proper dimensional alignments
- Human-constraint respecting designs

ENFORCEMENT:
- Pre-check before interface deployment
- Automated pattern validation
- User testing verification

HOLES:
- Need complete forbidden pattern catalog
- Need automated pattern checker
```

Block 3: DETECTION STACK

```
INTERFACE INTEGRITY DETECTORS:

Detector 1: Cognitive Load Analyzer
Input: Interface state
Output: U1.x level
Signals: Element count, grouping, nesting depth
Threshold: >15 ungrouped elements = overload warning

Detector 2: Interaction Type Classifier
Input: User actions
Output: U2.x classification
Signals: Action types (click, type, drag, etc.)

Detector 3: Response Time Monitor
Input: Action → Feedback latency
Output: U3.x classification
Signals: Measured milliseconds
```

Alerts: >1s for interactive elements

Detector 4: Error State Tracker
Input: Error conditions + handling
Output: U4.x classification
Signals: Error type, recovery options available

Detector 5: Learnability Assessor
Input: Time to task completion
Output: U5.x classification
Signals: New user success rate, time to proficiency

Detector 6: Navigation Path Tracker
Input: User navigation events
Output: U6.x classification + confusion metrics
Signals: Back presses, repeated navigation, abandonment

Detector 7: Visual Hierarchy Analyzer
Input: UI rendering
Output: U7.x classification
Signals: Size differences, color contrast, z-index

Detector 8: Paradigm Consistency Checker
Input: Interaction patterns
Output: U8.x classification + inconsistencies
Signals: Action types across interface

Detector 9: Personalization Level Detector
Input: Configuration options
Output: U9.x classification
Signals: Customization surface area

Detector 10: Pattern Coherence Validator
Input: Full signature [U1-U9]
Output: Valid/invalid pattern
Signals: Dimensional compatibility rules

Detector 11: User Frustration Monitor
Input: User behavior signals
Output: Frustration level
Signals: Error rate, retry attempts, rage clicks, abandonment

HOLES:
- Need baseline metrics for each dimension
- Need trained ML models for some detectors
- Need analytics instrumentation

Block 4: CONTAINMENT TOPOLOGY

```
INTERFACE ORGANIZATION MODULES:

Module A: Core Interaction (Highest Authority)
- Primary task flow
- Main actions
- Critical path
Cannot be compromised for secondary features

Module B: Secondary Features
- Supporting functionality
- Optional tools
- Moderate priority

Module C: Tertiary/Settings
- Configuration
- Advanced features
- Low priority

Module D: Error/Feedback
- Error messages
- Loading states
- Progress indicators

Module E: Help/Onboarding
- Tutorials
- Documentation
- First-time experience

Coupling Rules:
- Core → Secondary (visible but not intrusive)
- Secondary → Tertiary (hidden until needed)
- Error/Feedback → All (immediate, overlay)
- Help → Core (integrated for novices)

Firebreaks:
- Error in Module A → halt and fix immediately
- Complexity in Module C → acceptable (advanced users)
- Confusion in Module E → revise onboarding

HOLES:
- Need module priority system
- Need progressive disclosure strategy
```

Block 5: CORRECTION STACK

```
USABILITY REPAIR MOVES:

Repair 1: Cognitive Load Reduction
IF cognitive overload detected (U1.7-U1.8 or >15 ungrouped elements)
THEN:
  - Add grouping/chunking
  - Use progressive disclosure
  - Hide secondary features
  - Simplify visual design

Repair 2: Responsiveness Improvement
IF feedback too slow for interaction type
THEN:
  - Add loading indicators
  - Optimize performance
  - Use optimistic UI updates
  - Provide immediate acknowledgment

Repair 3: Error Handling Enhancement
IF errors are catastrophic or cryptic
THEN:
  - Add confirmation dialogs for destructive actions
  - Implement undo/redo
  - Improve error messages (actionable, clear)
  - Add validation earlier in flow

Repair 4: Learning Curve Smoothing
IF abandonment rate high OR time-to-proficiency excessive
THEN:
  - Add onboarding tutorial
  - Provide contextual help
  - Use progressive feature introduction
  - Simplify initial experience

Repair 5: Navigation Clarification
IF users lost OR excessive back-navigation
THEN:
  - Add breadcrumbs
  - Show current location
  - Reduce modal depth
  - Add search/jump-to functionality

Repair 6: Hierarchy Strengthening
IF important actions missed OR visual chaos
```

```
THEN:
  - Increase size/color contrast for primary actions
  - Add white space
  - Reduce visual noise
  - Use clear grouping

Repair 7: Consistency Restoration
IF mixed paradigms OR pattern violations
THEN:
  - Standardize interaction patterns
  - Use platform conventions
  - Add mode indicators if paradigm must change
  - Align with user expectations

Repair 8: Pattern Coherence Correction
IF dimensional mismatch detected
THEN:
  - Identify conflicting dimensions
  - Adjust to nearest valid pattern
  - Prioritize user type and task requirements

HOLES:
- Need automated repair suggestions
- Need A/B testing framework
- Need user behavior analytics
```

Block 6: RESERVED CAPACITY

```
INTERFACE BUDGETS:

Budget 1: Attention Budget
- Primary action: Gets 60-80% of visual weight
- Secondary actions: 20-30%
- Tertiary: 5-10%
- Exceeding = cognitive overload

Budget 2: Interaction Budget
- Core task: ≤3 clicks/taps to complete
- Secondary task: ≤5 clicks/taps
- Advanced: ≤10 clicks/taps
- Exceeding = abandonment risk

Budget 3: Response Time Budget
- Touch/click: <100ms feedback
- Page load: <1s
- Search: <3s
```

```
- Background task: Progress indicator + notification

Budget 4: Error Tolerance Budget
- 0 catastrophic errors allowed
- <5% error rate for novice users
- <1% for expert users
- Exceeding = trust loss

Budget 5: Learning Time Budget
- Consumer app: <5 minutes to first success
- Productivity tool: <1 hour to proficiency
- Professional tool: <1 week to productive
- Exceeding = abandonment

HOLES:
- Need calibration per app type
- Need user cohort segmentation
```

Block 7: META-UPDATE LOOP

```
INTERFACE EVOLUTION TRACKING:

Meta-Law 1: Pattern Effectiveness Learning
"Track which patterns succeed for which tasks/users"
"Build pattern-to-success-rate database"
"Update pattern recommendations"

Meta-Law 2: Anti-Pattern Detection
"Log all usability issues + root cause patterns"
"Identify recurring anti-patterns"
"Add to forbidden pattern list"

Meta-Law 3: New Pattern Discovery
"If novel successful patterns emerge, validate and add to catalog"
"Require: sustained success + user integration"

Meta-Law 4: Accessibility Expansion
"Track accessibility violations"
"Expand requirements as standards evolve"
"Ensure WCAG compliance"

HOLES:
- Need analytics pipeline
- Need pattern success metrics
- Need feedback loop to designers
```

Appendix A: THREAT MODEL

```
THREATS TO INTERFACE USABILITY:

T1: Feature Creep
- Adding features increases density
- Violates cognitive load limits
- Mitigation: Progressive disclosure, feature flagging

T2: Responsiveness Degradation
- Performance decay over time
- Violates feedback timing requirements
- Mitigation: Performance monitoring, budgets

T3: Error Handling Neglect
- Edge cases not handled
- Catastrophic errors exposed
- Mitigation: Comprehensive error mapping

T4: Inconsistency Accumulation
- Different designers/teams use different patterns
- Violates consistency law
- Mitigation: Design system, pattern library

T5: Accessibility Oversight
- Works for some, not others
- Violates universal usability
- Mitigation: Accessibility testing, WCAG compliance

T6: Mobile-Desktop Mismatch
- Pattern works on one, fails on other
- Device-specific violations
- Mitigation: Responsive design, adaptive patterns

T7: Expert-Bias
- Designers assume user knowledge
- Learning curve too steep
- Mitigation: Novice user testing

HOLES:
- Need threat mitigation playbook
```

Appendix B: FAILURE→RETURN MAP

```
USABILITY FAILURE → REPAIR MAPPING:
```

```
F1: Cognitive Overload
→ R1: Cognitive Load Reduction
   Simplify, group, hide secondary

F2: Slow Response
→ R2: Responsiveness Improvement
   Optimize, add loading states

F3: Catastrophic Errors
→ R3: Error Handling Enhancement
   Add confirmation, undo, better messages

F4: High Abandonment
→ R4: Learning Curve Smoothing
   Onboarding, help, simplification

F5: Navigation Confusion
→ R5: Navigation Clarification
   Breadcrumbs, location indicators

F6: Missed Primary Actions
→ R6: Hierarchy Strengthening
   Visual emphasis, reduce noise

F7: Inconsistent Experience
→ R7: Consistency Restoration
   Standardize patterns

F8: Pattern Violation
→ R8: Pattern Coherence Correction
   Align to valid pattern

F9: Accessibility Failure
→ R9: Accessibility Enhancement
   WCAG compliance, alternative interactions

HOLES:
- Need automated repair prioritization
```

LML: LAW INTEGRATION

LML Candidates:

✅ LAW UI.1: COGNITIVE CAPACITY

```
INTERFACE MUST NOT exceed human working memory capacity (7±2 items) for untrained

Scope: All consumer interfaces
Enforcement: Element count + grouping detector
Violation: → NSL.UI.COGNITIVE_LOAD_REDUCTION
```

✅ LAW UI.2: FEEDBACK RESPONSIVENESS

```
INTERACTIVE ELEMENTS MUST provide feedback within expected timeframe:
- Touch/click: <100ms
- Operations: <1s
- Complex tasks: Progress indicator required

Scope: All interactive interfaces
Enforcement: Response time monitor
Violation: → NSL.UI.RESPONSIVENESS_IMPROVEMENT
```

✅ LAW UI.3: ERROR RECOVERABILITY

```
DESTRUCTIVE/IRREVERSIBLE operations MUST have confirmation OR undo capability.

Scope: All interfaces with state changes
Enforcement: Error handling analyzer
Violation: → NSL.UI.ERROR_HANDLING_ENHANCEMENT
```

✅ LAW UI.4: LEARNABILITY APPROPRIATENESS

```
LEARNING CURVE MUST match target user type:
- Consumer: <5 min to first success
- Professional: <1 week to proficiency

Scope: All applications
Enforcement: Time-to-proficiency metrics
Violation: → NSL.UI.LEARNING_CURVE_SMOOTHING
```

✅ LAW UI.5: NAVIGATION TRANSPARENCY

```
USER MUST always know current location and have clear path to return.

Scope: All multi-screen interfaces
Enforcement: Navigation path tracker
Violation: → NSL.UI.NAVIGATION_CLARIFICATION
```

✅ LAW UI.6: VISUAL HIERARCHY REQUIREMENT

```
PRIMARY ACTIONS MUST be visually distinguishable from secondary/tertiary actions.

Scope: All interfaces with multiple actions
Enforcement: Visual hierarchy analyzer
Violation: → NSL.UI.HIERARCHY_STRENGTHENING
```

✅ LAW UI.7: INTERACTION CONSISTENCY

```
INTERACTION PARADIGM MUST be consistent throughout interface OR clearly mode-indi

Scope: All interfaces
Enforcement: Paradigm consistency checker
Violation: → NSL.UI.CONSISTENCY_RESTORATION
```

✅ LAW UI.8: DIMENSIONAL COHERENCE

```
ALL 9 DIMENSIONS MUST form valid, compatible pattern from 343 catalog.

Scope: All interfaces
Enforcement: Pattern coherence validator
Violation: → NSL.UI.PATTERN_COHERENCE_CORRECTION
```

✅ LAW UI.9: ACCESSIBILITY BASELINE

```
INTERFACE MUST meet WCAG 2.1 Level AA standards minimum.

Scope: All public-facing interfaces
Enforcement: Accessibility testing tools
Violation: → NSL.UI.ACCESSIBILITY_ENHANCEMENT
```

REJECTED (Not Laws):

❌ **Aesthetic Preferences** → Subjective

- "Flat design is better than skeuomorphic" = trend, not law

❌ **Feature Recommendations** → Context-dependent

- "Add dark mode" = nice-to-have, not requirement

❌ **Platform-Specific Guidelines** → Implementation detail

- “Use iOS Human Interface Guidelines” = standard, not universal law

---

NSL: REPAIR MOVES

NSL.UI.COGNITIVE_LOAD_REDUCTION

```
Trigger: >15 ungrouped elements OR user cognitive overload detected
Move: Simplify interface
Effect: Brings density within human capacity

Algorithm:
1. Identify element count and grouping
2. Apply progressive disclosure (hide secondary features)
3. Add visual grouping (cards, sections, white space)
4. Chunk related items
5. Verify ≤7 top-level groups
```

NSL.UI.RESPONSIVENESS_IMPROVEMENT

```
Trigger: Feedback delay exceeds threshold for interaction type
Move: Optimize or add loading states
Effect: Maintains user flow

Algorithm:
1. Measure actual latency
2. If <100ms achievable → optimize performance
3. If unavoidable delay → add loading indicator immediately
4. For >3s operations → add progress bar + time estimate
5. Verify perceived responsiveness
```

NSL.UI.ERROR_HANDLING_ENHANCEMENT

```
Trigger: Catastrophic error OR no recovery mechanism for destructive action
Move: Add confirmation and/or undo
Effect: Prevents irreversible mistakes

Algorithm:
1. Identify destructive operations
2. Add confirmation dialog (U4.2)
3. OR implement undo stack (U4.3)
4. Improve error messages (actionable, specific)
```

5. Verify error protection in place

NSL.UI.LEARNING_CURVE_SMOOTHING

Trigger: High abandonment OR time-to-proficiency exceeds budget
Move: Add onboarding and simplify
Effect: Reduces barrier to adoption

Algorithm:
1. Measure time-to-first-success
2. Add interactive tutorial for core tasks
3. Provide contextual help
4. Simplify initial experience (progressive disclosure)
5. Verify improved metrics

NSL.UI.NAVIGATION_CLARIFICATION

Trigger: User lost OR excessive back-navigation
Move: Add location indicators
Effect: User always knows where they are

Algorithm:
1. Detect navigation confusion (analytics)
2. Add breadcrumbs for hierarchical navigation
3. Show current location visually
4. Reduce modal depth if >3 layers
5. Add search/jump-to if deep hierarchy

NSL.UI.HIERARCHY_STRENGTHENING

Trigger: Primary actions missed OR visual chaos
Move: Emphasize important elements
Effect: Clear action priorities

Algorithm:
1. Identify primary/secondary/tertiary actions
2. Increase size/color contrast for primary
3. Add white space around important elements
4. Reduce visual weight of secondary items
5. Verify clear visual hierarchy

NSL.UI.CONSISTENCY_RESTORATION

```
Trigger: Mixed interaction paradigms OR pattern violations
Move: Standardize to single paradigm
Effect: Predictable interaction

Algorithm:
1. Detect paradigm inconsistencies
2. Choose primary paradigm (most common or platform-standard)
3. Convert inconsistent interactions
4. OR add clear mode indicators if paradigm must change
5. Verify consistency throughout
```

NSL.UI.PATTERN_COHERENCE_CORRECTION

```
Trigger: Dimensional mismatch in signature
Move: Adjust to nearest valid pattern
Effect: Coherent, usable pattern

Algorithm:
1. Identify incompatible dimensions
2. Find nearest valid pattern from 343 catalog
3. Prioritize user type and task requirements
4. Adjust conflicting dimensions minimally
5. Verify pattern validity
```

NSL.UI.ACCESSIBILITY_ENHANCEMENT

```
Trigger: WCAG violations detected
Move: Fix accessibility issues
Effect: Universal usability

Algorithm:
1. Run accessibility audit (automated tools)
2. Fix identified violations:
   - Add alt text for images
   - Ensure keyboard navigation
   - Improve color contrast
   - Add ARIA labels
3. Manual testing with assistive tech
4. Verify WCAG 2.1 AA compliance
```

DELTA: INTEGRATION RESULTS

LML_ADD:


```
[
  {"law_id": "UI.1", "text": "Cognitive capacity limit", "ledger": "UI.1.v1.2026-
  {"law_id": "UI.2", "text": "Feedback responsiveness", "ledger": "UI.2.v1.2026-0
  {"law_id": "UI.3", "text": "Error recoverability", "ledger": "UI.3.v1.2026-02-1
  {"law_id": "UI.4", "text": "Learnability appropriateness", "ledger": "UI.4.v1.2
  {"law_id": "UI.5", "text": "Navigation transparency", "ledger": "UI.5.v1.2026-0
  {"law_id": "UI.6", "text": "Visual hierarchy requirement", "ledger": "UI.6.v1.2
  {"law_id": "UI.7", "text": "Interaction consistency", "ledger": "UI.7.v1.2026-0
  {"law_id": "UI.8", "text": "Dimensional coherence", "ledger": "UI.8.v1.2026-02-
  {"law_id": "UI.9", "text": "Accessibility baseline", "ledger": "UI.9.v1.2026-02
]
```


NSL_ADD:


```
[
  {"repair_id": "NSL.UI.COGNITIVE_LOAD_REDUCTION", "maps_failures": ["F1"]},
  {"repair_id": "NSL.UI.RESPONSIVENESS_IMPROVEMENT", "maps_failures": ["F2"]},
  {"repair_id": "NSL.UI.ERROR_HANDLING_ENHANCEMENT", "maps_failures": ["F3"]},
  {"repair_id": "NSL.UI.LEARNING_CURVE_SMOOTHING", "maps_failures": ["F4"]},
  {"repair_id": "NSL.UI.NAVIGATION_CLARIFICATION", "maps_failures": ["F5"]},
  {"repair_id": "NSL.UI.HIERARCHY_STRENGTHENING", "maps_failures": ["F6"]},
  {"repair_id": "NSL.UI.CONSISTENCY_RESTORATION", "maps_failures": ["F7"]},
  {"repair_id": "NSL.UI.PATTERN_COHERENCE_CORRECTION", "maps_failures": ["F8"]},
  {"repair_id": "NSL.UI.ACCESSIBILITY_ENHANCEMENT", "maps_failures": ["F9"]}
]
```


DETECTION_SPEC:

- Cognitive Load Analyzer
- Response Time Monitor
- Error State Tracker
- Learnability Assessor
- Navigation Path Tracker
- Visual Hierarchy Analyzer
- Paradigm Consistency Checker
- Pattern Coherence Validator
- Accessibility Auditor
- User Frustration Monitor

BLOCKERS:

HOLES_BLOCKING_LML:

- Need baseline cognitive load metrics per user type
- Need response time requirements per interaction type
- Need learnability success criteria

HOLES_BLOCKING_NSL:

- Need automated UI analysis tools
- Need user behavior analytics
- Need A/B testing framework

SUMMARY

From EFUE collapse (343 UI/UX patterns):

✅ **9 LML Laws** (human-centered interface requirements)
✅ **9 NSL Repairs** (usability issue fixes)
✅ **10 Detection Systems** (usability monitoring)
✅ **Flagged Blocking Holes** (need metrics, analytics, testing)

Status: Laws + Repairs defined, ready for implementation.

COMPLETE: UI/UX → LML/NSL pipeline executed.

Infinite interface possibilities → 343 patterns → 9 laws + 9 repairs.